Name: ____________________ Tutorial group: ____________

Matriculation number: | | | | | | | | | |

26 September 2023 **MH1812 Test 1** 60 minutes

QUESTION 1. **(30 marks)**

Use mathematical induction to show that

$$1 \cdot 3 + 2 \cdot 4 + \cdots + (n-1)(n+1) = \frac{(n-1)n(2n+5)}{6}.$$

for all integers $n \geq 2$.

[Proof:]

Base case: when $n = 2$, the left-hand side is $1 \cdot 3 = 3$ and the right-hand side is $\frac{1 \cdot 2 \cdot 9}{6} = 3$. The equality holds.

Inductive step: Suppose that the equality holds for $n = k$, and we shall show that it holds for $n = k + 1$, that is, we shall show that

$$1 \cdot 3 + 2 \cdot 4 + \cdots + (k-1)(k+1) + k(k+2) = \frac{k(k+1)(2k+7)}{6}. \tag{1}$$

By induction hypothesis, the LHS of (1) equals

$$\begin{aligned} \frac{(k-1)k(2k+5)}{6} + k(k+2) &= k\left(\frac{2k^2+3k-5}{6} + \frac{6k+12}{6}\right) \\ &= k \cdot \frac{2k^2+9k+7}{6} \\ &= k \cdot \frac{(2k+7)(k+1)}{6}, \end{aligned}$$

which is exactly the RHS of (1).

Therefore, by mathematical induction, the original equality holds for all integers $n \geq 2$.

For graders only	Question	1	2(a)	2(b)	3(a)	3(b)	Bonus	Total
	Marks							

QUESTION 2. **(30 marks)**

(a) (10 marks) Find 202300000000^{1812} mod 9.

[Solution:] Note that

$$202300000000 \equiv 2023 \cdot 10^8 \equiv 7 \cdot 1 \equiv 7 \pmod 9,$$

the problem is reduced to 7^{1812} mod 9. Observe that $7^3 \equiv 1 \pmod 9$ and 1812 is divisible by 3, the answer is thus

$$1^{1812/3} \bmod 9 = (1 \bmod 9) \bmod 9 = 1.$$

(b) Let $\mathbb{R}^+$ denote the set of positive real numbers. For $x, y \in \mathbb{R}^+$, let $P(x, y)$ denote the predicate "$x - y$ is an integer". What are the truth values of these statements?

(i) (10 marks) $\forall x \in \mathbb{R}^+, \exists y \in \mathbb{R}^+, P(x, y)$.

(ii) (10 marks) $\exists x \in \mathbb{R}^+, \forall y \in \mathbb{R}^+, P(x, y)$.

[Solution:]

(i) The statement is true. For any $x \in \mathbb{R}^+$, let $y = x$, then $x - y = 0$ is an integer, i.e., $P(x, y)$ is true.

(ii) The statement is false. Let $x \in \mathbb{R}^+$. If x is not an even integer, let $y = x/2$, then $x - y = x/2$, which is not an integer; if x is an even integer, let $y = 1/2$, then $x - y$ is not an integer.

QUESTION 3. **(40 marks)**

(a) (20 marks) Determine whether $((\neg q \wedge r) \wedge (q \wedge s)) \to ((q \vee \neg r) \to s)$ is a tautology.

[Proof:] It is a tautology. No matter what values q takes, $q \wedge \neg q$ is false and thus the condition of the implication is false, so the whole sentence is true.

(b) (20 marks) Determine whether the following argument is valid. (Please state the names of the inference rules whenever you apply them.[1])

$$\begin{aligned} & q \vee r; \\ & t \to s; \\ & q \to \neg s; \\ & \neg r; \\ \therefore\ & \neg t. \end{aligned}$$

[Solution:] The argument is valid, as shown by the following inference table.

Step	Formula	Reason
(1)	$q \vee r$	Premise
(2)	$\neg r$	Premise
(3)	q	(1)+(2), disjunctive syllogism
(4)	$q \to \neg s$	Premise
(5)	$\neg s$	(3)+(4), modus ponens
(6)	$t \to s$	Premise
(7)	$\neg t$	(5)+(6), modus tollens

[1] The inference rules you may use are: modus ponens, modus tollens, conjunctive simplification, conjunctive addition, disjunctive addition, disjunctive syllogism, hypothetical syllogism and rule of contradiction.

BONUS QUESTION. **(10 marks)**

[Points will be given to *fully* correct solutions only. The total mark of this test is capped at 100 marks.]

Find all tuples of primes (p, q, r) such that

$$p(q - r) = q + r.$$

[Solution:] The given equation can be rewritten as

$$(p-1)q = (p+1)r, \tag{2}$$

or

$$q = \left(1 + \frac{2}{p-1}\right) r.$$

This implies that $(p-1)|2r$. Note that $2r$ has at most 4 divisors, 1, 2, r and $2r$. Thus $p-1$ must be one of them.

In $p - 1 = 1$, then $p = 2$ and then $q = 3r$, contradicting that q is a prime.

In $p - 1 = 2$, then $p = 3$ and then $q = 2r$, contradicting that q is a prime.

In $p - 1 = r$, since p and r are both primes, it must hold that $p = 3$ and $r = 2$. Then $q = 4$ is not a prime, contradiction.

In $p - 1 = 2r$, then $q = r + 1$. Since q and r are primes, it must hold that $q = 3$ and $r = 2$, so $p = 5$.

To conclude, the only tuple of primes satisfying the given equation is $p = 5$, $q = 3$ and $r = 2$.

[Alternative Solution:] It also follows from (2) that

$$\frac{q}{r} = \frac{p+1}{p-1}.$$

If $p = 2$, then $q = (p+1)r$, contradicting the assumption that q is a prime. Hence $p \geq 3$ and p is odd. We can then write

$$\frac{q}{r} = \frac{(p+1)/2}{(p-1)/2}.$$

Since q and r are primes, the left-hand side is in its most reduced form. For the right-hand side, note that $(p+1)/2$ and $(p-1)/2$ are consecutive integers, hence the right-hand side is also in its most reduced form. It must then hold that $q = (p+1)/2$ and $r = (p-1)/2$. This implies that $q = r + 1$ and so $r = 2$, $q = 3$, $p = 5$. This is the only solution.